SUPPORTING ADULT GUIDANCE

Finding a side

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner uses a trigonometric ratio to **find an unknown side** from one angle and one side, by **writing the ratio as an equation and solving it with algebra** — deliberately *not* by memorising a rule. Every solution is four lines: write the ratio, substitute (unknown = x), rearrange, evaluate. This lesson keeps the unknown in the numerator of the ratio, so the rearrangement is 'multiply both sides by the known number'. The emphasis is on **choosing the correct ratio**, understanding the **algebra** of the rearrangement, and reading the ratio value from a **calculator in degrees**. The unknown-in-the-denominator case (which needs a different rearrangement) is the next lesson.

Driving Question: *How do I find a length from one side and one angle?*

The mathematics, in plain terms

This is where trigonometry starts to do real work — and it is taught as **solving an equation**, not applying a formula. The reason matters: a learner who memorises 'unknown = known × ratio' has one more rule to confuse with others, whereas a learner who **writes the ratio as an equation and rearranges it** is using algebra they already understand, and the *same* thinking will handle the next lesson without a new rule. Every solution is the same four lines. Take finding the opposite when the hypotenuse is 8 and the angle is 40°: **(1)** write the ratio, sin 40° = opposite/hypotenuse; **(2)** put in the numbers with the unknown as x, sin 40° = x/8; **(3)** rearrange — x is divided by 8, so multiply both sides by 8, giving x = 8 × sin 40°; **(4)** evaluate on a calculator in degrees, x = 8 × 0.643 = 5.1 cm. The key teaching point is line 3: it is ordinary equation-solving, not a special trig move. Because this lesson keeps the unknown in the numerator, line 3 is always 'multiply both sides by the known number' — but framing it as algebra (rather than as a rule) is what lets the learner handle the unknown-in-the-denominator case next lesson by the same reasoning. Two things go wrong: the **wrong ratio** (a side that isn't in the problem) and the **wrong algebra** (dividing when the equation says multiply). And the calculator must be in **degrees**, not radians — a wrong mode is the commonest source of nonsense answers (check sin 30° = 0.5). The lesson ends with 'reach the unmeasurable' problems that show the point: one measurable distance and one angle give a height you could never reach.

Solve an equation, don't memorise a rule

The heart of this lesson is a deliberate pedagogical choice: find the side by **writing the ratio as an equation and solving it**, rather than by recalling 'unknown = known × ratio'. It is worth being explicit with the learner about *why* — remembering many separate techniques is exactly what causes confusion, whereas understanding the algebra means there is only ever one thing to do: write the equation and rearrange it. Insist on all four lines every time — write the ratio, substitute (unknown = x), rearrange, evaluate — even when the learner can see the answer, because the habit of showing the rearrangement is what makes the next lesson easy. Line 3 is the one to dwell on: 'x is divided by the known number, so multiply both sides by it'. That is plain equation-solving, and naming it as such (not as a trig rule) is the whole point. Keep the calculator in **degrees**. Finishing with real 'reach the unmeasurable' problems answers 'what is this for?' and makes the method memorable.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Set it up: choose the ratio, then reveal the four-line solution (ratio → substitute → rearrange → evaluate) one line at a time.	~10 min
Movement	Side Finder: choose the ratio, type the rearrangement yourself, then evaluate on a calculator — the working builds line by line.	~14 min
Reflection	Set up or slip?: judge two lines of working — right ratio, and correct algebra (multiply, not divide)?	~10 min
Creativity	Reach the unmeasurable: solve the ratio equation to find a real height from a distance and an angle.	~6 min
Rest	A pause after finding a length you could never reach.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — indirect measurement, and understanding over memorised rules.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Trying to memorise a rule instead of solving the equation

Repair: “Write the ratio as an equation, put in x, and rearrange. You already know how to solve equations — there's nothing new to remember.”

Wrong algebra in line 3 — dividing when x is in the numerator

Repair: “In line 2, x is divided by the known number, so what do you do to both sides to undo that? Multiply.”

Choosing the wrong ratio

Repair: “Which two sides are in this problem — the known one and the one you want? Write the ratio that uses exactly those two.”

Calculator left in radians, not degrees

Repair: “Check the mode — it must say DEG. sin 30° should give exactly 0.5; if it doesn't, you're in radians.”

Multiplying by the angle instead of the ratio of the angle

Repair: “You multiply by sin 40°, which is 0.643 — not by 40. Work out the ratio value, then evaluate.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the three ratios and SOH · CAH · TOA (Lesson 4), naming the sides relative to an angle (Lesson 2), reading a trig value from a calculator in degrees, and rearranging a simple equation (multiplying both sides). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner writing the four lines and reasoning through the algebra:

- “Which two sides are in this problem? So write the ratio as an equation.”
- “Put in the numbers with the unknown as x — what is line 2?”
- “x is divided by the known number. What do you do to both sides?”
- “Now evaluate on your calculator (in degrees) — what is x?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why not just teach unknown = known × ratio?”

Because it is one more rule to remember and confuse with others. Writing the ratio as an equation and rearranging uses algebra the learner already knows, and the same reasoning solves the next lesson (unknown in the denominator) without a new rule.

“What is the algebra in line 3, exactly?”

In line 2 the unknown x is divided by the known number (e.g. $x/8$). To get x on its own you do the inverse — multiply both sides by that number — giving $x = 8 \times \sin 40°$. It is ordinary equation-solving.

“My answer looks wildly wrong — what happened?”

Most often the calculator is in radians. Switch it to degrees and check that $\sin 30° = 0.5$. Also make sure you multiplied by the ratio value (a decimal), not by the angle itself.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.

Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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